# VSDBabySoC Physical Design and Multi-Corner Timing Analysis

OpenROAD Physical Design Flow and Timing Analysis Across SKY130 HD PVT Corners

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|---|---|
| **Assessment Date** | 10/10/2026 |
| **Design** | VSDBabySoC |
| **Top Module** | vsdbabysoc |
| **Physical Design / Timing Tool** | OpenROAD GUI / OpenROAD-flow-scripts |
| **Technology** | SkyWater SKY130 HD |
| **PVT Timing Corners** | 16 |

Project Report

# 1. Aim

To perform physical design analysis of the VSDBabySoC design using OpenROAD-flow-scripts with the SkyWater sky130hd technology and evaluate setup and hold timing across the supplied PVT corners.

# 2. Project Overview

This project examines the VSDBabySoC physical design flow, including floorplanning, standard-cell placement, Clock Tree Synthesis (CTS), routing, and Static Timing Analysis (STA). The supplied evidence includes a post-route OpenROAD GUI view and timing summaries for 16 PVT corners. The timing graphs are based on the values visible in the supplied terminal screenshot.

# 3. Objectives

• Understand the major physical design stages of VSDBabySoC.

• Inspect the routed layout and timing-report interface in OpenROAD GUI.

• Compare worst setup slack and worst hold slack across the supplied PVT corners.

• Identify corners with negative setup slack that require investigation.

• Document the measured values without inventing or replacing report data.

# 4. Tools and Technologies

• OpenROAD-flow-scripts and OpenROAD GUI

• SkyWater sky130hd platform

• GitHub Codespaces and noVNC

• Linux terminal and Tcl commands

• Static Timing Analysis (STA) and PVT timing libraries

# 5. Theory

## 5.1 Physical Design Flow

**Floorplanning:** defines the core and die regions, rows, tracks, macro locations, and power planning. **Placement:** assigns legal locations to standard cells. **CTS:** builds a clock distribution network. **Routing:** connects cells through metal layers and vias. Post-route parasitic estimation improves the realism of timing analysis.

## 5.2 Static Timing Analysis

STA checks timing paths using timing constraints and cell/interconnect delay models rather than functional simulation. Setup analysis checks whether data arrives in time for capture; hold analysis checks whether data remains stable long enough after the clock edge. For both checks, non-negative slack indicates that the reported check meets its timing requirement; negative slack indicates a violation.

## 5.3 PVT Corners

PVT represents Process, Voltage, and Temperature. Different process models, supply voltages, and temperatures change cell and interconnect delay. Multi-corner analysis helps reveal conditions under which setup or hold timing is most critical.

# 6. Procedure Summary

• **Launch the environment:** open the GitHub Codespace, connect through noVNC, and start the OpenROAD GUI or terminal.

• **Inspect physical stages:** load the appropriate stage database (ODB), matching SDC constraints, and sky130hd RC configuration.

• **Inspect post-route layout:** review metal layers, vias, signal routing, and the timing report panel.

• **Collect timing results:** extract the worst setup (max) and hold (min) slack from each corner report.

• **Compare corners:** plot setup and hold slack using the measured report values and flag negative setup slack.

Illustrative command pattern (replace paths with files actually present in the run directory):

```
read_db ./results/sky130hd/vsdbabysoc/base/5_route.odb
read_sdc ./results/sky130hd/vsdbabysoc/base/5_route.sdc
source ./platforms/sky130hd/setRC.tcl
report_checks -path_delay max
report_checks -path_delay min
```

The exact database filenames and paths depend on the flow configuration. Load the matching database, SDC, libraries, and parasitics before interpreting timing results.

# 7. Multi-Corner Timing Results

The table below transcribes the 16 corner values shown in the supplied terminal screenshot. Slack values are in nanoseconds (ns). “Worst slack (max)” is used here as the setup-slack metric and “worst slack (min)” as the hold-slack metric, consistent with the screenshot labels.

| PVT corner | Setup slack (ns) | Hold slack (ns) |
|---|---|---|
| ff_100C_1v65 | 8.15 | 0.25 |
| ff_100C_1v95 | 9.03 | 0.20 |
| ff_n40C_1v56 | 6.91 | 0.29 |
| ff_n40C_1v65 | 7.62 | 0.25 |
| ff_n40C_1v76 | 8.24 | 0.23 |
| ff_n40C_1v95 | 8.94 | 0.19 |
| ss_100C_1v40 | -2.72 | 0.89 |
| ss_100C_1v60 | 2.10 | 0.63 |
| ss_n40C_1v28 | -29.05 | 1.68 |
| ss_n40C_1v35 | -15.57 | 1.25 |
| ss_n40C_1v40 | -9.98 | 1.05 |
| ss_n40C_1v44 | -6.71 | 0.94 |
| ss_n40C_1v60 | 0.43 | 0.63 |
| ss_n40C_1v76 | 3.73 | 0.48 |
| tt_025C_1v80 | 7.13 | 0.31 |
| tt_100C_1v80 | 7.26 | 0.31 |

**Setup observation:** the most negative setup slack is –29.05 ns at ss_n40C_1v28. Other negative setup-slack corners are ss_n40C_1v35 (–15.57 ns), ss_n40C_1v40 (–9.98 ns), ss_n40C_1v44 (–6.71 ns), and ss_100C_1v40 (–2.72 ns). These indicate setup timing violations in the reported results.

**Hold observation:** all 16 listed hold-slack values are positive, ranging from 0.19 ns to 1.68 ns. The smallest listed hold slack is 0.19 ns at ff_n40C_1v95. This suggests the reported worst hold checks meet timing, subject to confirming that each report was generated with the correct constraints and parasitics.

# 8. Timing Graphs

The following graphs were supplied for this project. They visually compare the worst hold and setup slack across the 16 PVT corners.

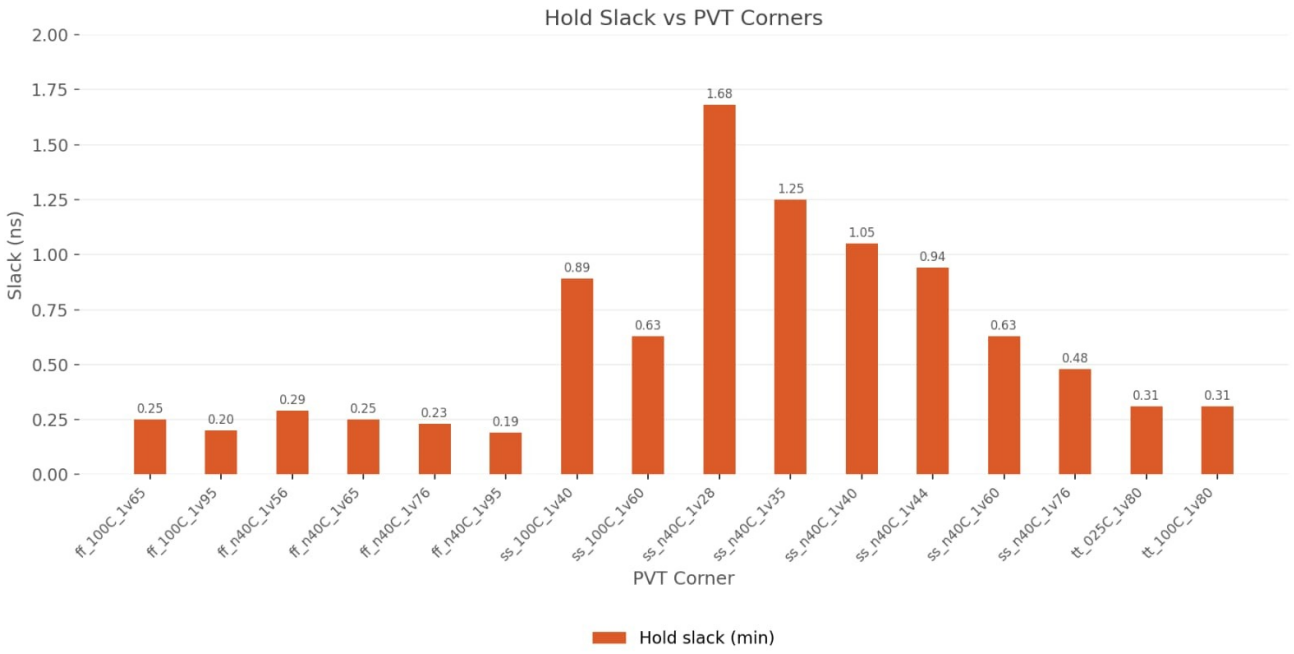



Figure 1. Hold slack versus PVT corners (source: supplied graph).

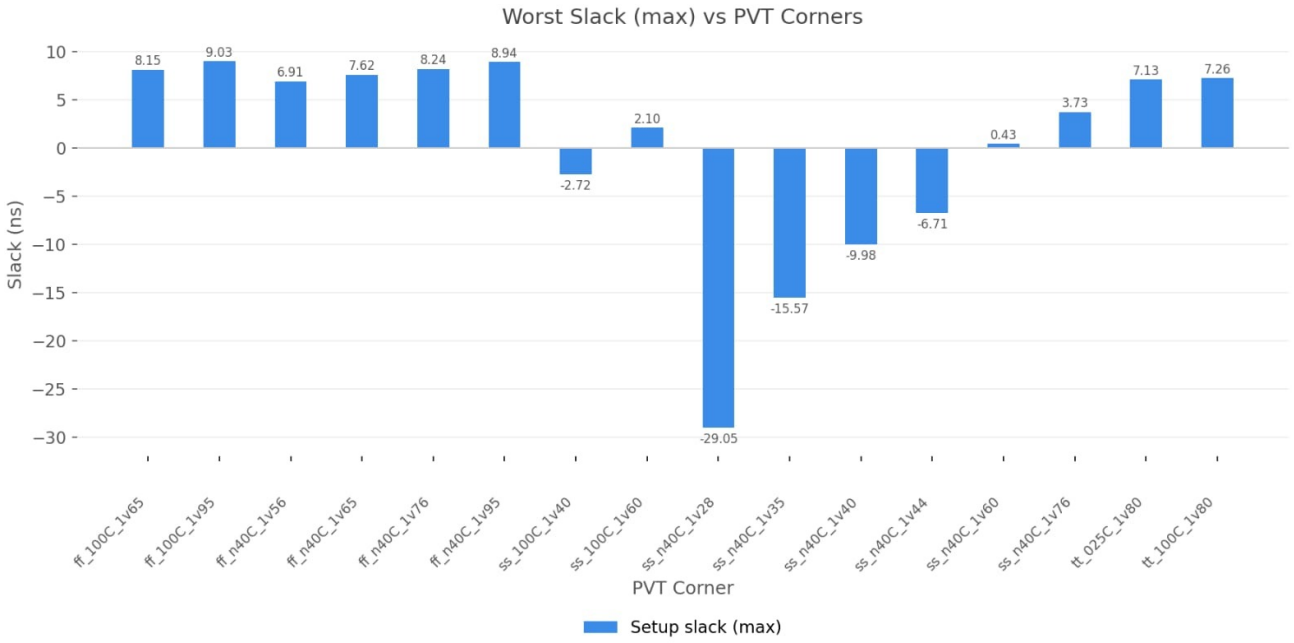



Figure 2. Worst setup slack versus PVT corners (source: supplied graph).

# 9. OpenROAD GUI and Terminal Evidence

The screenshots below document the timing summary extracted from the corner reports and the OpenROAD post-route GUI view.

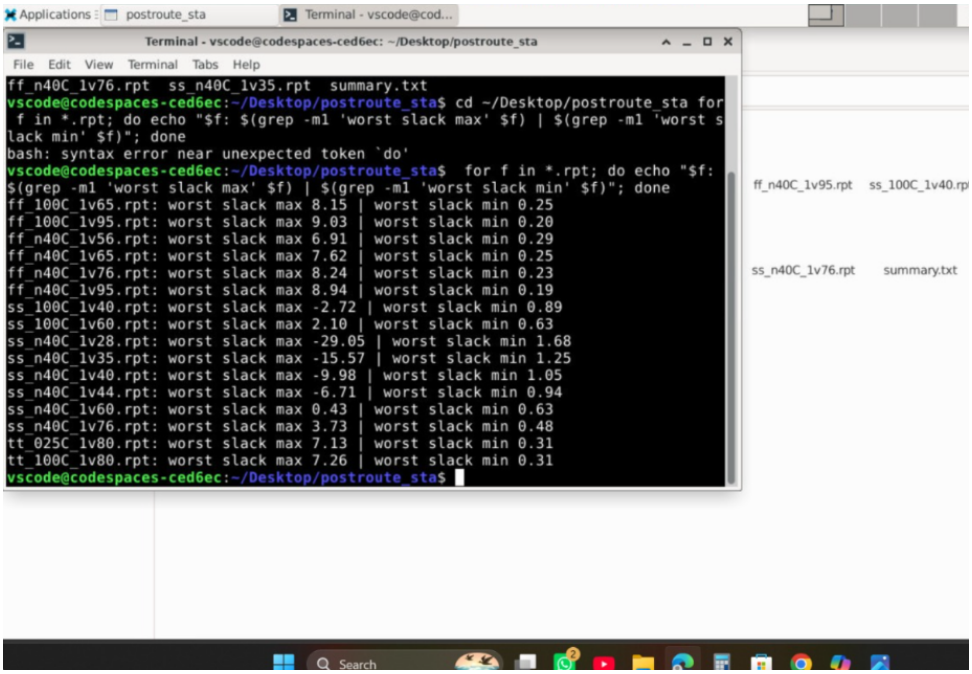



Figure 3. Terminal output showing setup and hold slack values for the 16 supplied corner reports.

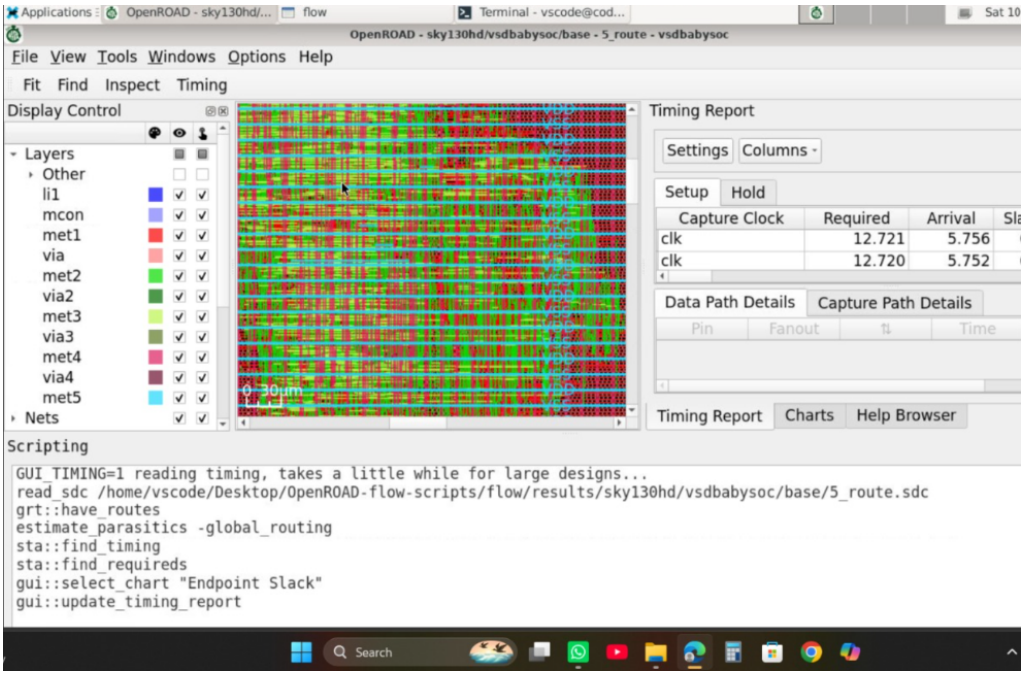



Figure 4. OpenROAD GUI showing the routed VSDBabySoC layout and timing report interface.

# 10. Discussion and Possible Improvements

• Investigate the most negative setup-slack paths first, beginning with ss_n40C_1v28.

• Inspect the startpoint, endpoint, cell delay, net delay, clock arrival, and required arrival time for each failing path.

• Check that each corner uses the intended timing libraries, voltage/temperature conditions, SDC constraints, and parasitic model.

• Potential remedies may include cell sizing, buffering, placement/routing improvements, or logic restructuring, but changes should be made only after identifying the actual critical path and confirming the design constraints.

• Re-run timing analysis after any change and verify both setup and hold timing.

## 11. Conclusion

The supplied VSDBabySoC results demonstrate how post-route static timing can be compared across 16 PVT corners using OpenROAD and the sky130hd platform. The recorded hold slack values are positive, while five corners show negative setup slack. The most severe setup result is –29.05 ns at ss_n40C_1v28, making that corner the first candidate for critical-path investigation. Final timing closure should be claimed only after validating the reports, constraints, and parasitics and then correcting and rechecking any violations.

## 12. Notes on Result Interpretation

This report is based on the values visible in the supplied screenshots; it does not independently rerun the OpenROAD flow. The screenshot contains 16 corner entries. If the assignment requires a different set of 16 PVT corners, update the table and graphs from the actual corresponding report files. The reported values should be verified against the raw reports before final submission.